UP Special Topic 5 — Agriculture, Agro-Climatic Zones, Irrigation Networks, Rural Economy and Research Institutes

Current Affairs

Year	Fact	Why it matters
Rolling	Refresh edition-bound figures from latest Census / Survey / State CA desks.	Do not invent interim totals.

Teaching cards

2.1 Agrarian Profile, Cropping Patterns and Major Crops of Uttar Pradesh

A. Land Utilization & Cropping Intensity

- **Total Geographical Area**: 240.93 lakh hectares; **Net Sown Area**: ~165.7 lakh hectares (~68.7% of total area).
- **Cropping Intensity**: ~160%–165%, indicating extensive double-cropping facilitated by pervasive tube-well irrigation and canal networks.
- **Three Core Cropping Seasons**: 1. **Kharif Season (Sowing: June–July; Harvesting: Sept–October)**:
  - Dependent on the South-West Monsoon.
  - Major Crops: **Paddy (Rice), Maize, Jowar, Bajra, Arhar (Tur), Urad, Moong, Groundnut, Soybean, Sesame (Til)**. 2. **Rabi Season (Sowing: October–November; Harvesting: March–April)**:
  - Dependent on winter soil moisture and Western Disturbance rains.
  - Major Crops: **Wheat, Barley, Gram (Chana), Pea (Matar), Lentil (Masoor), Mustard (Sarson), Potato, Linseed (Alsi)**. 3. **Zaid Season (Sowing: March–April; Harvesting: May–June)**:
  - Short summer window requiring assured irrigation.
  - Major Crops: **Watermelon, Muskmelon, Cucumber, Gourds, Moong, Sunflower, Fodder maize**.

B. Detailed Crop Geography

1. **Wheat**: - Uttar Pradesh is the **largest producer of wheat in India**, contributing over **32%–35%** of national wheat output. - Requires 10^\circ ext{C} - 15^\circ ext{C} during sowing/vegetative growth and 20^\circ ext{C} - 25^\circ ext{C} during grain filling/harvesting, with 50–75 cm rainfall. - Major producing districts: **Meerut, Muzaffarnagar, Bulandshahr, Moradabad, Bareilly, Badaun, Aligarh, Gorakhpur**.
2. **Sugarcane**: - Cash crop occupying ~22–25 lakh hectares; UP is the **#1 producer of sugarcane** in India. - Primary concentration in two belts:
   - *Western UP Belt*: Meerut, Muzaffarnagar, Saharanpur, Bijnor, Moradabad, Baghpat, Bulandshahr.

- *Terai / Eastern Belt*: Deoria, Kushinagar, Gorakhpur, Lakhimpur Kheri, Basti, Gonda.
- *Exam Distinction*: UP produces the highest **gross volume** of cane, but Maharashtra typically records higher **sucrose recovery percentage and yield per hectare** due to tropical maritime climatic advantages.

3. **Paddy (Rice)**: - Primary kharif staple occupying ~60 lakh hectares. - Densely cultivated in the high-rainfall **Terai and Eastern Plains (Purvanchal)**: Siddharthnagar, Maharajganj, Kushinagar, Gorakhpur, Deoria, Varanasi, Chandauli (known as the **"Rice Bowl of UP"**). - **Kalanamak Rice**: Cultivated in Siddharthnagar, Balrampur, and Basti; prized for its rich aroma, black husk, and high iron-zinc content (GI-tagged).
4. **Potato**: - UP ranks **#1 in India** in potato production (~35% of national output). - Major potato belt: **Agra, Firozabad, Farrukhabad, Kannauj, Mainpuri, Etah, Aligarh, Hathras, Mathura**. - International research hub: **CIP-SARC located at Singna, Agra**.
5. **Pulses and Oilseeds**: - Pulses (Gram, Lentil, Arhar) are heavily concentrated in the rainfed **Bundelkhand region** (Hamirpur, Banda, Jalaun, Jhansi, Mahoba, Lalitpur). - Mustard and Rapeseed dominate in **Western UP and South-Western semi-arid districts** (Agra, Mathura, Aligarh).
6. **Mentha (Peppermint)**: - UP accounts for over **80% of India's total mentha oil production** (India is the world's largest exporter of natural menthol). - Major clusters: **Barabanki, Badaun, Rampur, Sambhal, Moradabad, Bareilly, Jalaun**. Supported by CSIR-CIMAP, Lucknow.

2.2 Horticulture and Fruit Belts of Uttar Pradesh

HORTICULTURE SPECIALIZATION IN UP		
Fruit / Product	Premier District / Cluster	Iconic Variety
1. Mango	Malihabad (Lucknow), Saharanpur, Bulandshahr, Varanasi	Dasheri, Chausa, Langra, Safeda
2. Guava	Prayagraj (Allahabad), Kaushambi	Allahabad Surkha, Safeda, Apple Colour
3. Aonla	Pratapgarh (Largest producer)	Banarasi, Chakaiya, Francis (Hathijhool)
4. Citrus / Sweet Orange	Saharanpur (Orange/Lemon), Bundelkhand	Mosambi, Malta, Kagzi Lime
5. Betel Leaf (Paan)	Mahoba, Lalitpur, Banda, Varanasi	Mahoba Desi, Bangla, Maghai (GI Tagged)

- **Agro-Export Zones (AEZ) in UP**:
- *Mango and Vegetables AEZ*: Lucknow (Malihabad), Unnao, Hardoi, Sitapur, Barabanki.
- *Mango AEZ*: Saharanpur, Muzaffarnagar, Meerut, Bulandshahr, Baghpat.
- *Potato AEZ*: Agra, Hathras, Farrukhabad, Kannauj, Meerut, Aligarh, Baghpat.
- *Basmati Rice AEZ*: Western UP districts (Meerut, Muzaffarnagar, Saharanpur, Bareilly, Rampur).

2.3 Irrigation Architecture and Groundwater Dynamics

A. Source-Wise Net Irrigated Area Breakdown

According to official UP statistical land and water data:

SOURCES OF IRRIGATION IN UTTAR PRADESH	
Irrigation Source	**Percentage Share of Net Irrigated Area**
1. Tube-wells (Nalkoop)	~74.9% (Overwhelmingly Dominant)
• Private Tube-wells	(~71.2%)
• Government Tube-wells	(~3.7%)
2. Canals (Nahar)	~15.2%
3. Traditional Open Wells	~8.8%
4. Tanks, Lakes & Ponds	~0.5%
5. Other Sources	~0.6%

B. Groundwater Extraction & "Dark Zones"

- UP has the largest replenishable groundwater resources in India (~**40.7 billion cubic metres - bcm**).
- However, heavy concentration of private electric/diesel tube-wells in **Western UP** (Meerut, Saharanpur, Muzaffarnagar, Baghpat, Aligarh) has led to severe over-exploitation, creating notified **Over-Exploited Blocks ("Dark Zones")** and critical water table depletion.
- State interventions: **Mukhyamantri Khet Talab Yojana** (farm ponds in Bundelkhand), promotion of micro-irrigation (drip/sprinkler), and solar pump energization under PM-KUSUM.

2.4 Major State Agricultural Universities (SAUs)

1. **Chandra Shekhar Azad University of Agriculture & Technology (CSAUAT), Kanpur**: - Established in **1975** (evolved from the Government Agricultural College, Kanpur, founded 1906).
2. **Acharya Narendra Deva University of Agriculture & Technology (ANDUAT), Kumarganj (Ayodhya)**: - Established in **1975**; caters to the agro-development of eastern UP.
3. **Sardar Vallabhbhai Patel University of Agriculture & Technology (SVPUAT), Modipuram (Meerut)**: - Established in **2000**; caters to the agricultural needs of the Western Plain.
4. **Banda University of Agriculture & Technology (BUAT), Banda**: - Established in **2010**; dedicated to dryland farming and pulses research in the Bundelkhand region.
5. **Sam Higginbottom University of Agriculture, Technology and Sciences (SHUATS), Naini (Prayagraj)**: - Established in **1910** by Dr. Sam Higginbottom as Allahabad Agricultural Institute (now a premier private deemed university).
6. **Rani Lakshmi Bai Central Agricultural University (RLBCAU), Jhansi**: - Established in **2014** by an Act of Parliament as a **Central Agricultural University** for the Bundelkhand region.

Bilingual Terminology -- UP Agriculture

English	Hindi	Key Anchor
Kharif Crops	खरीफ फसल	Sown in June-July; harvested Oct-Nov (rice, maize, bajra)
Rabi Crops	रबी फसल	Sown in Nov; harvested Mar-Apr (wheat, mustard, gram)
Zaid Crops	जायद फसल	Short summer season between Rabi and Kharif (watermelon, cucumber)
Canal Irrigation	नहर सिंचाई	Upper/Lower Ganga Canal; Sharda Canal; Agra Canal
Tube Well	नलकूप	Main irrigation method in UP (groundwater); GBIC
Drip Irrigation	टपक सिंचाई	Efficient; promoted under PMKSY
Sardar Sarovar	सरदार सरोवर परियोजना	Not in UP; Gujarat; included for context
Sardar Vallabhbhai Patel Gau Samvardhan	गोवंश संवर्धन	UP cow protection policy
Pradhan Mantri Fasal Bima Yojana	प्रधानमंत्री फसल बीमा योजना	Crop insurance scheme (2016)
e-Nam	राष्ट्रीय कृषि बाजार	National Agriculture Market; online trading platform
MSP	न्यूनतम समर्थन मूल्य	Minimum Support Price; recommended by CACP
Cooperative Farming	सहकारी कृषि	Joint farming by cooperative societies
Zaid Crops	जायद	Summer crops
GI Tag	भौगोलिक संकेत	Geographical Indication Tag protecting origin-specific products
Green Revolution	हरित क्रांति	1960s; HYV seeds (Norman Borlaug/M.S. Swaminathan)

Extended Theory -- UP Agriculture: Key Data and Canal Systems

UP's Agricultural Significance

UP is the **backbone of Indian food security**: - Largest producer of **Sugarcane** (48%+ of national production) - Largest producer of **Potato** (30%+ of national production) - Largest producer of **Milk** (~17% of national production) - 2nd largest producer of **Wheat** (after MP, some years Punjab) - Major producer of **Mustard, Gram, Lentil, Paddy**

```
        UP MAJOR IRRIGATION CANALS
        ____________________________________________
```

```
UPPER GANGA CANAL (1854)
└ Source: Haridwar barrage (River Ganga)
└ Irrigates: Western UP (Saharanpur to Kanpur division)
└ Oldest major irrigation project in British India

LOWER GANGA CANAL (1878)
└ Offtake from Upper Ganga Canal at Narora Barrage
└ Irrigates: Bulandshahr, Aligarh, Etah, Mainpuri

SHARDA CANAL (1928)
└ Source: Sharda (Kali) River at Banbasa, Champawat (UK)
└ Irrigates: Eastern UP (Lakhimpur, Sitapur, Hardoi, Unnao)

AGRA CANAL (1874)
└ Source: Yamuna River at Okhla Barrage
└ Irrigates: Agra and Mathura districts

RIHAND (GOVIND BALLABH PANT SAGAR) (1962)
└ Source: Rihand River
└ Irrigation + Power (Obra, Rihand Thermal)
└ Largest reservoir in India by area
```

UP Agricultural Seasons

Season	Crop Examples	Sowing	Harvesting
Kharif (खरीफ)	Paddy (rice), Maize, Bajra, Sugarcane, Soybean	June-July (SW monsoon)	October-November
Rabi (रबी)	Wheat, Mustard, Gram, Masoor, Barley	October-November	March-April
Zaid (जायद)	Watermelon, Cucumber, Moong Dal	March-April	June-July

One-Liner Revision -- UP Agriculture

#	Fast Fact
1	UP = largest producer of **Sugarcane** in India
2	UP = largest producer of **Potato**
3	UP = largest producer of **Milk**
4	UP has the **most sugar mills** in India (100+)
5	Upper Ganga Canal (1854) = oldest major irrigation work; Haridwar barrage
6	Sharda Canal (1928) = eastern UP irrigation from Kali River
7	Rihand Dam reservoir = **Govind Ballabh Pant Sagar** (largest by area)

#	Fast Fact
8	Kharif crops sown in **June-July**; harvested in **Oct-Nov**
9	Rabi crops sown in **Oct-Nov**; harvested in **Mar-Apr**
10	MSP recommended by **CACP** (Commission for Agri Costs and Prices)
11	PMFBY (2016) = Pradhan Mantri Fasal Bima Yojana (crop insurance)
12	e-NAM = National Agriculture Market (online APMC platform)
13	Green Revolution (1960s) = HYV seeds; M.S. Swaminathan led in India
14	UP groundwater irrigation = Tube wells (dominant method)
15	GI Tag: UP products = Banarasi silk, Chikankari, Aligarh locks